Controlled Vocabulary Derived from Prototype Quantity Taxonomy v20260402

AI-generated from data and documents of three METIS pilot projects with tabular data

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An **Angle** is a geometric entity that

- Consists of two rays sharing a common endpoint (the vertex)
- Is defined by the amount of rotation required to superimpose one ray onto the other
- Is measured in units such as degrees, radians, or gradians
- Quantifies planar separation between intersecting lines

Angular Measurements quantify rotational displacement, orientation, and angular position of objects, instruments, and measurement configurations, expressed in degrees or radians.

An **Anisotropy Field Uncertainty** is a measurement uncertainty that

- Pertains to the magnetic anisotropy field parameter of a material
- Originates from limitations of the measurement instrumentation and calibration procedures
- Reflects variability introduced by sample heterogeneity and environmental conditions
- Is expressed using statistical metrics such as standard deviation, confidence interval, or expanded uncertainty

An **Anisotropy Field** is a magnetic field that

- Originates from the material's magnetic anisotropy energy
- Represents the effective field needed to overcome anisotropy and align magnetization along a hard direction
- Is proportional to the anisotropy constant divided by the saturation magnetization
- Is expressed in tesla (or oersted) and influences ferromagnetic resonance behavior

An **Applied Magnetic Field** is a magnetic field that

- Is generated by an external source to influence a material or device
- Is oriented and controlled to achieve a specific magnitude and direction
- Is spatially uniform or configured as required for the experiment or measurement.

An **Attempt Frequency** is a rate parameter that

- Quantifies the number of vibrational attempts per unit time to overcome an energy barrier

- Appears as the pre-exponential factor in Arrhenius-type expressions for transition rates
- Has units of frequency (s^{-1})
- Is determined by the vibrational modes of the system at the transition state

An **Average** is a statistical measure that

- Aggregates a set of numerical values
- Is calculated by dividing the sum of the values by the count of values
- Represents the central tendency of the data set
- Is affected by each value in the data set
- Includes variants such as arithmetic, geometric, and harmonic

A **Background** is a contextual element that

- Describes the prior state or conditions surrounding a measurement
- Includes information about sources of noise, interference, or baseline signals
- Documents experimental setup, sample history, and environmental parameters
- Supports interpretation, comparison, and reproducibility of data
- Is recorded alongside primary measurement results

Bandwidth is a communication parameter that

- Quantifies the maximum rate of data transfer across a transmission medium
- Is expressed in bits per second (bps) or hertz (Hz)
- Determines the capacity for simultaneous data streams
- Is constrained by physical media characteristics and signaling protocols

A **Beam Power Standard Deviation** is a statistical metric that

- Quantifies the dispersion of measured beam power values around their mean
- Is calculated as the square root of the variance of a set of beam power measurements
- Has the same units as beam power (e.g., watts)
- Is derived from repeated power measurements performed under consistent experimental conditions
- Provides an indication of beam output stability and repeatability

A **Beam Power** is a physical quantity that

- Quantifies the rate of energy transport carried by a beam of particles or electromagnetic radiation
- Is expressed in units of watts ($J \cdot s^{-1}$)
- Equals the product of beam intensity (energy per unit area per unit time) and the beam's cross-sectional area

- Serves as a key performance metric for lasers, particle accelerators, and other beam-based instruments

A **Beam Time** is a scheduled research facility resource that

- Designates a specific time interval during which a user operates a particle accelerator or beamline
- Is allocated through a proposal review and scheduling process
- Provides access to the beam's particles, photons, or neutrons for data collection

Calibrated refers to an instrument condition that

- Has been adjusted to align its output with known reference standards
- Exhibits documented traceability to national or international standards
- Has undergone verification procedures establishing accuracy within specified tolerances
- Includes recorded calibration data and associated uncertainty estimates

A **Camera Angle** is a photographic parameter that

- Specifies the orientation of the camera sensor relative to the subject's spatial position
- Determines the perspective by altering the apparent depth and scale of objects
- Influences narrative tone through high, low, or eye-level positioning
- Establishes the viewer's line of sight toward the subject

A **Camera Rotation Angle** is an angular measurement that

- Quantifies the rotational displacement of a camera around a specified axis
- Is expressed in degrees or radians
- Defines the camera's orientation relative to a reference coordinate system
- Is used to transform captured image coordinates into world coordinates

A **Damping Constant Uncertainty** is a measurement uncertainty that

- Pertains to the damping constant parameter of a dynamic system
- Is expressed as a statistical interval, standard deviation, or confidence bound
- Originates from limitations in measurement instrumentation, calibration, or data analysis
- Contributes to the overall uncertainty budget in model predictions or performance assessments

A **Damping Constant** is a system parameter that

- Quantifies the proportional relationship between damping force and velocity in a damped system
- Possesses units of $force \cdot time$ per length (e.g., $N \cdot s/m$) or mass per time
- Remains constant for a given linear viscous damping model

A **Damping Factor Uncertainty** is a measurement uncertainty that

- Quantifies the deviation between the true damping factor of a system and its reported value
- Arises from limitations of measurement instruments, environmental variations, and data analysis methods
- Is expressed as a statistical interval or confidence level
- Affects the reliability of dynamic response predictions

A **Damping Factor** is a dimensionless parameter that

- Quantifies the rate at which oscillations in a dynamic system decrease over time
- Is defined as the ratio of actual damping to critical damping in a second-order system
- Determines whether a system is under-damped, critically damped, or over-damped

A **Damping Parameter Uncertainty** is a parameter uncertainty that

- Quantifies the lack of exact knowledge about the numerical value of a damping coefficient used in device or process models
- Arises from variations in material properties, fabrication tolerances, and measurement noise
- Is expressed using statistical descriptors such as standard deviation, confidence interval, probability distribution
- Impacts the reliability of simulation outcomes and prediction of device performance

A **Damping Parameter** is a model parameter that

- Quantifies the rate at which oscillatory energy is dissipated in a dynamic system
- Is expressed as a scalar (or dimensionless) value representing viscous or structural damping
- Appears in equations of motion to modify the system's natural frequency and decay rate
- Is derived from material properties, geometry, and operating conditions
- Influences the amplitude attenuation of transient and steady-state responses

ΔH^0 **Uncertainty** is a measurement uncertainty that

- Quantifies the dispersion of repeated Δh_0 values obtained under identical conditions
- Derives from instrument noise, calibration error, and environmental variability affecting Δh_0 measurements
- Is expressed using statistical metrics such as standard deviation, variance, or confidence interval

Δh_0 is a metrology parameter that

- Quantifies the deviation of measured film thickness from the nominal zero-reference baseline
- Is expressed in units of length such as nanometers or angstroms
- Is calculated from calibration measurements using reference standards
- Reflects systematic offset in thickness measurement equipment

ΔH^0 is a standard thermodynamic quantity that

- Represents the enthalpy change of a chemical or physical process measured under defined standard conditions (1 bar pressure, specified temperature, pure substances in their reference states)
- Is expressed in units of energy per amount of substance (e.g., $kJ \cdot mol^{-1}$)
- Is obtained from heat exchange at constant pressure while the system and surroundings remain in equilibrium.

ΔH° **Uncertainty** is a measurement uncertainty that

- Quantifies the possible deviation between the reported standard enthalpy change (ΔH°) and its true value
- Is expressed as a numeric value with units of energy per amount of substance (e.g., $kJ \cdot mol^{-1}$)
- Results from propagation of uncertainties in experimental data and reference standards
- Accompanies ΔH° values in thermochemical databases and literature.

Dependent refers to an entity that

- Relies on another entity for support
- Has its state determined by another entity
- Lacks autonomy
- Is contingent upon another

A **Diffraction Efficiency Standard Deviation** is a statistical measure that

- Quantifies the variability of measured diffraction efficiency values across repeated experiments
- Equals the square root of the variance of diffraction efficiency values
- Is expressed in the same units as diffraction efficiency (e.g., percentage points)
- Is calculated from a sample set obtained under consistent measurement conditions.

A **Diffraction Efficiency** is an optical performance metric that

- Quantifies the fraction of incident optical power diffracted into a specified diffraction order
- Is expressed as a dimensionless ratio or percentage
- Is determined under defined wavelength, polarization, and incident-angle conditions

Dimensional and Geometric Properties quantify spatial extent, shape, orientation, and periodic distances in physical systems and measurement configurations.

A **Dimensional Property** is a property that

- Quantify the size or extent of an object in one or more spatial dimensions
- Are expressed as numeric values with associated measurement units
- Derive from geometric relationships such as length, area, volume, or angle
- Remain invariant under changes in material composition while varying with scale

Dimensional refers to a measurement attribute that

- Quantifies the number of independent spatial directions of an object or system
- Expresses size, extent, or scale along one or more axes
- Is represented by units such as length, area, volume, or higher-dimensional analogues
- Influences analysis, modeling, and visualization in metrology and semiconductor engineering

An **Effective Field** is a field that

- Represents the net influence of all microscopic interactions on a given degree of freedom
- Is derived by averaging or integrating out other degrees of freedom in a many-body system
- Replaces explicit many-body interactions in equations of motion or Hamiltonians
- Is spatially uniform or varies with position
- Determines the dynamics of the subject through the same formalism as a physical field
- Is used in mean-field and coarse-grained models to simplify complex interactions

An **Effective Magnetization Uncertainty** is a measurement uncertainty that

- Quantifies the estimated deviation between the reported effective magnetization value and the true value
- Is expressed as a statistical confidence interval or standard deviation
- Incorporates contributions from instrument calibration, sample geometry, temperature stability, and data analysis procedures
- Is derived using uncertainty propagation methods applied to the measurement model for effective magnetization

An **Effective Permittivity** is a material property that

- Represents the averaged dielectric response of a heterogeneous medium to an applied electric field

- Quantifies the combined influence of constituent permittivities and their spatial distribution
- Is used in modeling wave propagation and capacitance in composite or structured materials

Effective refers to a property that

- Achieves intended outcomes
- Produces measurable results
- Operates with efficiency

Electromagnetic and Network Metrics characterize electrical signal response, circuit behavior, and electromagnetic wave propagation.

Electromagnetic Response Properties quantify how materials and networks respond to electromagnetic signals, including permittivity, scattering parameters, and signal components.

An **Energy Barrier** is a potential energy feature that

- Appears as a local maximum on a potential energy surface
- Separates two adjacent minima representing distinct stable configurations
- Quantifies the minimum energy required to convert one configuration into the other
- Is measured in units of energy such as electronvolts or joules
- Originates from interatomic or intermolecular interactions that oppose the transition

Exchange is a magnetic material property that

- Quantifies the quantum mechanical interaction between neighboring magnetic moments in a material
- Arises from electron spin-spin coupling and orbital overlap
- Is expressed as an exchange energy or exchange constant (typically in units of joules or electron volts)
- Determines whether magnetic moments align parallel (ferromagnetism) or antiparallel (antiferromagnetism)
- Governs magnetic ordering, Curie temperature, and spin wave behavior

Experimental Context and Metadata provide information about how, when, and where data was collected, including setup parameters, timing, and bibliographic references.

An **Exposure Time** is a process parameter that

- Quantifies the duration that a target material is subjected to a specified radiation source
- Is expressed in units of time such as seconds, milliseconds, or microseconds
- Directly determines the total radiant energy delivered to the material
- Is set by the control system of the exposure equipment
- Influences the degree of photochemical or physical change in the material

A **Final Power** is a power measurement that

- Represents the ultimate output power of a device under test after completion of the test sequence
- Is recorded in units of watts with traceable calibration
- Corresponds to the specified test condition set (e.g., voltage, temperature, load)
- Is stored as a numeric value in the metrology data repository

A **First Order Diffraction Efficiency** is a diffraction efficiency metric that

- Quantifies the proportion of incident optical power diffracted into the first diffraction order of a grating
- Is expressed as a dimensionless ratio or percentage of the incident power
- Is measured under specified illumination wavelength, polarization, and incident angle conditions
- Depends on the grating's groove profile, material refractive index, and thickness

A **Frequency** is a physical quantity that

- Quantifies the number of occurrences of a repeating event per unit time
- Is expressed in units of hertz (Hz), representing cycles per second
- Characterizes oscillations, vibrations, electromagnetic waves, and periodic signals
- Is related to wavelength by the equation f = c / λ for electromagnetic radiation
- Determines the energy of photons via $E = h \cdot f$ in quantum mechanics

A **G Factor Uncertainty** is a measurement uncertainty that

- Quantifies the doubt associated with the determined value of the Landé g-factor for a semiconductor or microelectronic device
- Is expressed as a statistical interval (e.g., standard deviation or expanded uncertainty) in the same units as the g-factor
- Derives from contributions such as instrument precision, calibration error, and repeatability of the measurement
- Is reported together with the confidence level or coverage factor used in its evaluation

A **G Factor** is a dimensionless physical parameter that

- Quantifies the proportionality between magnetic moment and angular momentum of a particle, atom, or defect
- Is expressed as a unitless number
- Varies with the specific particle, electronic state, or crystalline environment

A **Gilbert Damping Parameter Uncertainty** is a parameter uncertainty that

- Pertains to the Gilbert damping constant, a material-specific coefficient describing energy dissipation in magnetization dynamics
- Expresses the quantitative range or statistical distribution associated with the measured or modeled value
- Originates from experimental measurement error, instrument calibration limits, and model fitting procedures
- Is reported using standard uncertainty metrics such as standard deviation, confidence interval, or covariance matrix

A **Gilbert Damping Parameter** is a material parameter that

- Quantifies the rate at which the precessional motion of magnetization dissipates energy into the lattice
- Appears as the dimensionless coefficient α in the Landau-Lifshitz-Gilbert equation
- Is intrinsic to ferromagnetic materials and influences the linewidth of ferromagnetic resonance

Gilbert Damping is a phenomenological damping parameter in magnetization dynamics that

- Quantifies the rate at which the precessional motion of magnetization loses energy
- Appears as a dimensionless coefficient (α) in the Landau-Lifshitz-Gilbert equation
- Acts as a torque opposing the magnetization vector
- Originates from spin-orbit coupling and other relaxation mechanisms
- Influences ferromagnetic resonance linewidth and magnetic switching speed

A **Grazing Angle of Incidence** is an angle of incidence that

- Is measured between the incoming radiation (or particle) direction and the surface plane
- Has a value close to zero degrees, i.e., the beam travels nearly parallel to the surface
- Is defined as the complement to the angle measured from the surface normal
- Is employed in techniques such as X-ray reflectivity and surface scattering to enhance surface sensitivity

A **Gyromagnetic Ratio Uncertainty** is a measurement uncertainty that
pertains to the gyromagnetic ratio, a physical constant linking magnetic moment and angular momentum
expresses the quantitative estimate of doubt about the reported value
is derived from statistical analysis of repeated measurements and assessment of systematic influences
is reported using standard uncertainty notation such as standard deviation or confidence interval

A **Gyromagnetic Ratio** is a physical constant that

- Relates a particle's magnetic dipole moment to its angular momentum
- Is expressed in units of coulomb per kilogram ($C \cdot kg^{-1}$) or $rad \cdot s^{-1} \cdot T^{-1}$
- Serves as a fundamental parameter in techniques such as nuclear magnetic resonance and electron spin resonance

A **Harmonic Number** is an experimental parameter that

- Identifies the order of a harmonic in diffraction, frequency generation, or nonlinear optical measurements
- Is expressed as a positive integer (1st harmonic, 2nd harmonic, 3rd harmonic, etc.)
- Specifies which diffraction order or harmonic mode is being measured or analyzed
- Is recorded as a part of the experimental configuration

The **Imaginary Part of S21** is a complex scattering-parameter component that

- Corresponds to the coefficient of the imaginary unit in the complex representation of S21
- Quantifies the reactive (phase-related) contribution to the forward transmission coefficient
- Is dimensionless and expressed as a pure number without units

Impedance is an electrical property that

- Relates voltage and current in alternating-current (AC) circuits through magnitude and phase
- Is expressed as a complex quantity composed of resistance (real part) and reactance (imaginary part)
- Has units of ohms (Ω)
- Varies with signal frequency

Independent refers to an attribute that

- Does not rely on external control or support
- Operates autonomously
- Exhibits self-sufficiency

An **Inhomogeneous Linewidth Broadening Uncertainty** is a measurement uncertainty that

- Arises from spatial variations in material properties that cause nonuniform spectral line widths
- Is quantified as the statistical dispersion of linewidth values across a sample population
- Is expressed as a confidence interval or standard deviation associated with the measured linewidth

- Is distinguished from homogeneous broadening uncertainty which originates from intrinsic processes

Inhomogeneous Linewidth Broadening is a spectral line broadening phenomenon that

- Originates from static variations in local environments of emitters within an ensemble
- Produces a distribution of transition frequencies across the population
- Yields a composite line shape reflecting the sum of individual narrow lines
- Is independent of the intrinsic decoherence time of individual emitters
- Is observed in materials with spatially varying strain, composition, or electric/magnetic fields

An **Intensity Standard Deviation** is a statistical measure that

- Quantifies the dispersion of intensity values around their mean
- Is computed as the square root of the variance of the intensity data
- Is expressed in the same units as the original intensity measurements
- Provides an indicator of consistency or repeatability in metrology results

An **Intensity** is a physical quantity that

- Quantifies the magnitude of a phenomenon per unit area or per unit time
- Is expressed in standardized units such as watts per square meter or decibels
- Represents the strength or power of a signal, field, or radiation

Line Width in a Diffraction Measurement is a measurement parameter that

- Quantifies the full width at half maximum (FWHM) of a diffraction peak
- Reflects the spread of diffracted intensity in reciprocal space
- Is derived from the angular or reciprocal-space profile of the diffracted beam
- Correlates with crystallite size, lattice strain, and instrumental broadening

A **Line Width** is a dimensional parameter that

- Denotes the lateral dimension of a patterned line feature on a semiconductor substrate
- Is measured perpendicular to the line's longitudinal axis
- Is expressed in nanometers or micrometers
- Serves as a critical dimension influencing device performance
- Is obtained using metrology techniques such as CD-SEM, scatterometry, or optical critical-dimension inspection

Linear Dimensions and Extents quantify the physical size, width, length, and spatial extent of features, structures, and components in one or more directions.

Linewidth Error is a measurement error that

- Quantifies the deviation between the intended or nominal line width of a patterned feature and the actual measured line width
- Originates from instrument resolution limits, process variations, and data-analysis algorithms
- Is expressed in nanometers or as a percentage of the target dimension
- Influences device performance by altering electrical characteristics

A **Linewidth Uncertainty** is a measurement uncertainty that

- Pertains to the estimated range of values for the measured spectral linewidth
- Is expressed as a standard deviation or confidence interval
- Derives from instrument resolution, noise, and fitting algorithm contributions
- Is combined with other uncertainty components according to metrology guidelines
- Is reported alongside the measured linewidth value

A **Spectral Linewidth** is a spectral property that

- Quantifies the range of wavelengths (or frequencies) over which a spectral line maintains significant intensity
- Is expressed as a width measurement such as full width at half maximum (FWHM)
- Reflects broadening mechanisms including natural lifetime, Doppler, pressure, and instrumental effects
- Influences resolution and discrimination capability in spectroscopic analysis

A **Magnetic Field Offset Uncertainty** is an uncertainty component that

- Represents the statistical dispersion of repeated offset measurements
- Originates from sensor calibration error, drift, and environmental magnetic interference
- Is expressed as a standard deviation or confidence interval around the estimated offset value
- Contributes to the total uncertainty budget of magnetic field measurements

A **Magnetic Field Offset** is a magnetic field parameter that

- Represents the constant deviation of a magnetic field sensor's output from the true zero-field condition
- Is determined through calibration against a known reference field
- Is expressed in units of magnetic flux density (e.g., tesla or gauss)
- Is applied as a correction factor to raw sensor data to obtain accurate field measurements

Magnetic Field Quantities represent vector or scalar fields that describe magnetic forces, interactions, and resonance phenomena in materials and devices.

Magnetic Material Properties characterize the intrinsic magnetic behavior of materials, including magnetization states, exchange interactions, and gyromagnetic responses.

A **Magnetic Relaxation Rate** is a relaxation rate that

- Quantifies the exponential decay of magnetization toward equilibrium after perturbation
- Is expressed in reciprocal time units (e.g., s^{-1})
- Depends on interactions such as spin–lattice or spin–spin coupling
- Is measured using techniques such as nuclear magnetic resonance or electron spin resonance

Magnetization is a magnetic property that

- Quantifies the magnetic dipole moment per unit volume of a material
- Results from the alignment of atomic or molecular magnetic moments within the material
- Is represented as a vector field indicating magnitude and direction
- Is measured in amperes per meter ($A \cdot m^{-1}$) in the SI system
- Reflects the material's response to an applied magnetic field.

A **Maximum Count** is a statistical measure that

- Represents the greatest integer value observed within a defined set of items or events
- Is determined by enumerating all occurrences and selecting the highest total
- Is expressed as a non-negative integer

A **Mean Beam Power** is a power metric that

- Represents the average optical power delivered by a beam over a defined time interval
- Equals total optical energy emitted divided by the duration of that interval
- Is expressed in units of watts (W)
- Applies to both continuous-wave and pulsed laser operation
- Excludes instantaneous power variations, focusing on the time-averaged value

A **Mean Beam Time** is a statistical metric that

- Quantifies the average duration of beam exposure per experiment
- Is calculated by dividing the total beam exposure time by the number of exposures
- Is expressed in units such as seconds, minutes, or hours
- Supports assessment of facility utilization and scheduling efficiency

A **Mean Beam** is a beam property that

- Quantifies the arithmetic average of measured intensity values across the spatial extent of the beam

- Is calculated by integrating intensity over the beam area and dividing by that area
- Provides a single representative value for comparing beam uniformity and power distribution

A **Mean Scan** is a data product that

- Aggregates multiple individual scan measurements of a semiconductor or microelectronic sample
- Computes the arithmetic average of corresponding data points across the scans
- Yields a single composite image or dataset representing the central tendency of surface topography or electrical characteristics
- Preserves the spatial resolution and coordinate system of the original scans
- Serves as a reference baseline for comparative metrology analysis

Measured refers to a quantified attribute that

- Has been obtained using a calibrated instrument
- Represents a numeric value
- Corresponds to a defined unit of measurement
- Is recorded in a data set

Measurement Errors represent the deviation between a measured value and its true or reference value, arising from systematic or random sources.

Measurement Uncertainties quantify the lack of exact knowledge about the true value of a measured quantity, expressed as statistical intervals, standard deviations, or confidence bounds.

Nominal refers to a reference value that

- Represents the intended or design specification for a parameter
- Provides a baseline for comparison with measured values
- Is expressed in the same units as the corresponding measured quantity
- Is used in tolerance specifications to define acceptable deviation

The **Number of Angles** is a geometric property that

- Quantifies the count of interior angles in a shape or polygon
- Is expressed as an integer representing the total angles present

The **Number of Beam Angles** is a metrology metric that

- Quantifies the count of distinct angular orientations a measurement beam assumes during data acquisition
- Is expressed as an integer value
- Influences the angular sampling resolution of the measurement system
- Is recorded as part of the experimental configuration metadata

The **Number of Beam Frames** is a quantitative metric that

- Represents the total count of discrete image frames captured during a beam-based measurement
- Is expressed as an integer value
- Increments by one each time a new frame is acquired
- Is recorded for each measurement session or experiment
- Provides information about data volume and temporal resolution of the measurement

The **Number of Frames** is a quantitative metric that

- Represents the count of discrete image frames acquired during a metrology operation
- Is expressed as an integer value
- Informs assessment of data volume and temporal resolution

Optical and Radiative Properties quantify light-matter interactions, diffraction phenomena, and electromagnetic radiation power and intensity.

Patterned refers to a visual attribute that

- Exhibits a repeated motif or arrangement
- Displays regular or systematic variation in form, color, or texture
- Conveys recognizable order or structure

Periodic Structures and Wavelengths characterize repeating spatial patterns, periodic distances, wave properties, and the geometric configuration of periodic systems.

Perpendicular Anisotropy is a magnetic anisotropy that

- Aligns the easy axis of magnetization orthogonal to the substrate or film plane
- Arises from interfacial, crystalline, or shape contributions in thin-film ferromagnetic layers
- Enhances out-of-plane magnetic stability for nanoscale elements
- Enables high coercivity and low switching currents in spin-torque devices

Physical Quantities and Constants encompass intrinsic properties of materials and fundamental physics parameters that characterize the behavior and state of physical systems.

Pitch in a Diffraction Measurement is a metrological parameter that

- Quantifies the distance between successive repeating units of a periodic structure
- Is derived from the angular positions of diffraction orders using the grating equation
- Is expressed in length units such as nanometers or micrometers
- Provides a direct measure of feature size for semiconductor and microelectronics components

Pitch is a geometric parameter that

- Quantifies the distance between adjacent repeating features in a periodic structure
- Is measured perpendicular to the line or edge of the repeating elements
- Is expressed in length units such as nanometers, micrometers, or millimeters
- Characterizes gratings, line arrays, and patterned structures in semiconductor and optical devices

Quality and Provenance Attributes describe the validation state, origin, and qualitative characteristics that establish the trustworthiness and context of measurement data.

A **Real Impedance** is a component of impedance that

- Represents the resistive part of the complex impedance
- Has a magnitude expressed in ohms without an associated phase angle
- Corresponds to power dissipation as heat in a circuit element

A **Real Part of S21** is a numeric attribute of a scattering parameter that

- Represents the real component of the complex transmission coefficient S21 in a two-port network
- Quantifies the in-phase amplitude of the signal transferred from port 1 to port 2
- Is expressed as a dimensionless ratio (or converted to dB) without imaginary contribution
- Is obtained from measured or simulated S-parameter data

A **Reference Impedance** is an impedance standard that

- Defines the nominal value against which circuit impedances are compared
- Is expressed in ohms
- Normalizes S-parameter measurements for network analyzers
- Matches the characteristic impedance of the associated transmission medium
- Remains fixed for a given measurement configuration

A **Reference** is a foundational benchmark for measurement that

- Provides a standardized value or material to define a specific magnitude of a physical quantity
- Links the measured data to an external authoritative standard (such as the SI) to ensure traceability
- Establishes the baseline or "zero point" (e.g., 0 Pa for absolute pressure) to provide context for the data

A **Reflection Coefficient** is a coefficient that

- Quantifies the ratio of the reflected wave amplitude to the incident wave amplitude at an interface
- Is expressed as a complex number representing both magnitude and phase

- Has a magnitude whose square equals the reflected power ratio
- Depends on the impedance mismatch between adjoining media

A **Resistance** is an electrical property that

- Quantifies opposition to the flow of electric charge in a conductor
- Is measured in ohms (Ω)
- Is proportional to the conductor's length and material resistivity
- Is inversely proportional to the conductor's cross-sectional area
- Follows Ohm's law linking voltage, current, and resistance ($\mathrm{V} = \mathrm{I}\,\mathrm{R}$)

An **S-Parameter** is a network parameter that

- Relates incident and reflected microwave waves at the ports of a linear electrical network
- Is expressed as a complex, frequency-dependent matrix of scattering coefficients
- Is measured under linear, time-invariant, small-signal conditions
- Characterizes transmission and reflection behavior of high-frequency components

A **Scattering Parameter** is a network parameter that

- Relates incident and reflected microwave signal wave amplitudes at each port of a linear, time-invariant electrical network
- Is expressed as a complex-valued matrix S whose elements are ratios of reflected to incident wave amplitudes
- Assumes a defined reference impedance for all ports
- Is measured with a vector network analyzer under small-signal conditions

A **Simulated** is a representation that

- Reproduces the observable behavior of a target system or phenomenon
- Is generated by a computational or mathematical model rather than by direct measurement
- Lacks physical embodiment of the original system
- Serves as input for analysis, design, or validation activities
- Operates under defined boundary conditions and parameters

Spectral Line Characteristics quantify the width, broadening, and shape of spectral features arising from wavelength or frequency distributions in optical and resonance measurements.

A **Spectral Linewidth** is a spectral property that

- Quantifies the range of wavelengths (or frequencies) over which a spectral line maintains significant intensity
- Is expressed as a width measurement such as full width at half maximum (FWHM)
- Reflects broadening mechanisms including natural lifetime, Doppler, pressure, and instrumental effects

- Influences resolution and discrimination capability in spectroscopic analysis

Stage Rotation is a positioning operation that

- Rotates a substrate or sample about a specified axis
- Provides precise angular displacement measured in degrees or radians
- Is driven by high-resolution actuators such as piezoelectric or stepper motors
- Maintains alignment of the sample with the measurement probe or optical system
- Is integrated into the coordinate system of a metrology instrument

Statistical Dispersions quantify the variability or spread of measurement values around a central tendency, typically expressed as standard deviations or variance.

A **Temperature** is a physical quantity that

- Quantifies the average kinetic energy of particles in a system
- Determines the direction of spontaneous heat transfer between systems
- Is measured in units such as kelvin, Celsius, or Fahrenheit
- Varies continuously with changes in energy content

Thermal Stability is a material property that

- Describes the ability of a material to retain its physical and chemical characteristics when exposed to elevated temperatures
- Resists degradation, phase change, or loss of functionality under thermal stress
- Maintains dimensional integrity and performance within a specified temperature range

Thermodynamic Quantities describe energy states, thermal properties, and enthalpy changes that govern the stability and phase behavior of systems.

A **Time Difference** is a temporal measurement that
  * quantifies the interval between two distinct timestamps
  * is expressed in standard time units (e.g., seconds, minutes, hours)
  * results from subtracting the earlier timestamp from the later timestamp

Traceable refers to a property that

- Has a documented lineage of data provenance
- Allows verification of origin through records
- Supports reproducibility by linking to source information

A **Transmission Coefficient** is a physical quantity that

- Expresses the ratio of transmitted wave amplitude (or power) to incident wave amplitude (or power) across a boundary
- Is dimensionless

- Is derived from the continuity conditions of the wave field at the interface
- Depends on frequency, material properties, and angle of incidence

Uncertainty, Error, and Quality Attributes quantify the doubt, variability, validation state, and provenance associated with measurements and data.

Validated is a quality-assurance status that an entity

- Underwent systematic verification against defined criteria
- Demonstrated compliance with applicable standards
- Established traceability to reference standards
- Provides documented evidence supporting its reliability

A **Waveguide Geometry** is a geometric configuration of a waveguide that

- Defines the cross-sectional shape and dimensions of the guiding structure
- Establishes the spatial distribution of material boundaries and dielectric properties
- Determines the supported electromagnetic modes and cutoff frequencies
- Influences propagation loss, dispersion, and confinement characteristics
- Includes bends, tapers, or periodic features that modify mode evolution

A **Waveguide Length** is a geometric dimension of a waveguide that

- Extends along the direction of signal propagation
- Is measured in linear units such as meters or millimeters
- Determines the electrical phase shift experienced by a wave traveling through the guide
- Is defined for a contiguous segment of the waveguide without discontinuities
- Influences insertion loss and group delay in a predictable manner

A **Waveguide Width** is a geometric dimension of a waveguide that

- Specifies the transverse physical extent of the waveguide cross-section
- Determines the cutoff frequency for modes propagating in the waveguide
- Is measured in micrometres or millimetres

A **Wavelength** is a wave property that

- Represents the spatial period of a periodic wave
- Equals the distance between consecutive points of identical phase (e.g., crest to crest)
- Is measured in units of length such as meters or nanometers
- Is mathematically related to frequency by the equation $\lambda = \mathrm{v} \,/\, \mathrm{f}$, where v is the wave speed
- Remains constant for a given wave frequency in a homogeneous medium

A **Write Error Rate** is a reliability metric that

- Quantifies the frequency of incorrect data written to a memory cell during write operations

- Is expressed as the number of write errors per total write attempts
- Is measured under specified test conditions and operating parameters
- Is used to assess memory device endurance and data integrity

An **X Bin** is a data bin that

- Groups measurement values of the variable X into discrete intervals
- Assigns each measurement to a specific interval based on its numeric range
- Facilitates statistical analysis and reporting of X-related metrology data

A **Y Bin** is a binning category that

- Used to group semiconductor devices based on measured electrical performance parameters
- Assigned during wafer-level test to indicate compliance with specification limits
- Recorded in metrology data sets for statistical analysis of yield